

SLM-125M — Completed Build Pipeline

From-scratch 125.8M-param Llama-style legal / financial small language model · 12L / 768d / 12h · 16,384 vocab · 1,024 context as of 2026-09-06

1 · DATA PIPELINE

Phases 0–4 · CPU · ~\$0.18

Smoke test + per-source token-yield measurement

Stream & deterministic clean — 718K → 698K docs kept

MinHash / LSH near-dedup + exact dedup + CaseHOLD/LexGLUE decontamination → ~670K docs

Train fresh 16,384 byte-level BPE tokenizer on the corpus

Tokenize + pack into 1024-token uint16 windows, 99 / 1 train-val split

RESULT

2.04B train tokens

1,991,282 train windows

realized mix: SEC 42% / case-law 35% /
FineWeb-Edu 23%

2 · PRETRAINING

Phase 5 · 8×H100 DDP (Modal) · ~\$8–9

1 epoch · 3,877 steps · ~2.03B tokens seen · 524K-token global batch

AdamW (β 0.9/0.95, wd 0.1), LR $6e-4 \rightarrow 6e-5$ cosine, 200M-token warmup

Completed in 2 runs — hit \$10 budget cap at step 3064, resumed from checkpoint

RESULT

train loss 9.107 → 2.4832

3 · BASE MODEL DEPLOY

Phase 6

Push base model + tokenizer to the Hugging Face Hub

RESULT

HF: IndraniBera/slm-125m-base (private)

4 · QA SUPERVISED FINE-TUNING

Phases 7–10 · 1×L4

Generate 1,196 QA pairs with gpt-5.4-nano, grounded in pretrain-corpus excerpts

Domain split: Legal 656 / SEC 300 / General 240

Tokenize into loss-masked, padded SFT examples

SFT-train from the base checkpoint · 2 epochs · LR $5e-5$

Base-vs-SFT head-to-head on CaseHOLD (held out of pretraining)

RESULT

assistant-token perplexity 8.11

CaseHOLD accNorm 0.170 → 0.166

HF: IndraniBera/slm-125m-sft (private)

5 · RAFT (retrieval-aware fine-tuning)

Phases 11–12 · 1×L4 · ~\$0.14

Reformat the same 1,196 pairs with 3 retrieved-style passages each

1 golden + 2 distractors ~79% of the time · 3 distractors, no golden ~21%

golden-present 942 / golden-absent 254

Tokenize RAFT examples (mirrors the SFT tokenizer exactly)

RAFT-train from the SFT checkpoint · 4 epochs · 296 steps · LR 3e-5

RESULT

assistant-token perplexity 2.821

HF: [IndraniBera/slm-125m-raft](#) (private)

6 · RAG INDEX

Phase 13 · CPU

Build BM25 lexical inverted index (in-memory, no vector DB)

6,000 chunks — legal 3,300 / SEC 1,500 / general 1,200

~880-char paragraph-snapped chunker (same noise profile as RAFT training)

RESULT

42,127 unique terms

retrieve k = 3

7 · FRONTEND & SERVING

Vercel + Modal FastAPI endpoints

Next.js app deployed on Vercel (project slm-125m)

/base-model completion playground + architecture / corpus / loss-curve viz — LIVE

/sft QA playground (SFT model) + dataset & eval stats — built locally, deploy pending

/rag RAG playground — RAFT model + live BM25 retrieval — built locally, deploy pending

Modal inference endpoints: Playground (live) · PlaygroundSFT · PlaygroundRAG

RESULT

LIVE: <https://slm-125m.vercel.app>

base playground deployed; /sft + /rag not yet pushed

NOT YET STARTED

RLAIF on the QA-SFT model — planned only (see 'instruction-QA SFT + RLAIF'). No reward-model / DPO / preference-data code exists yet.